Pricing Theory — Assignment 1

Covering Lectures 1–2

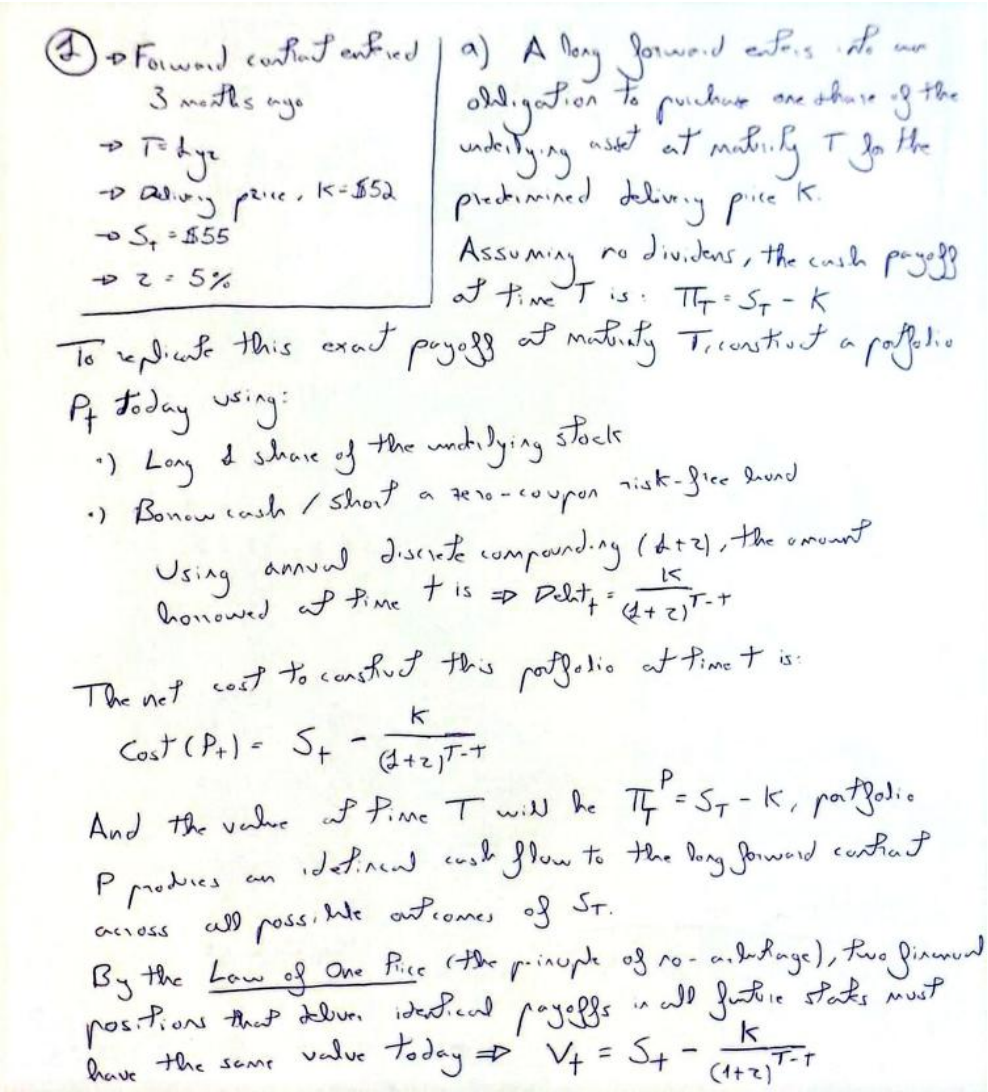
Question 1 — Valuing a Forward Contract After Inception

A long forward contract on a stock was entered into 3 months ago with T=1yr, with a delivery price of K = \$52. There are now 9 months remaining until maturity. The stock currently trades at S_t = \$55, and the risk-free rate is r = 5% per year (simple annual compounding). We assume no dividend.

(a) Derive a general expression for the value V_t of a long forward position at an arbitrary time t before maturity T, in terms of S_t, K, r, and the remaining time (T − t). Describe explicitly the portfolio you would hold today that replicates the contract's payoff at maturity (i.e. a replication argument.)

(b) Using your formula from (a), compute the numeric value of this specific forward position today.

(c) A counterparty offers to buy this forward position from you for \$6.00 today. Using your answer to (b), determine whether this price is fair. If it is not, construct an explicit, self-financing arbitrage strategy that locks in a riskless profit. Then repeat the analysis for a hypothetical offer of \$4.00 and explain how your strategy changes direction.



1 → Forward contract entered 3 months ago
→ T = 1yr
→ delivery price, K = \$52
→ S_t = \$55
→ r = 5%

a) A long forward entails the obligation to purchase one share of the underlying asset at maturity T for the predetermined delivery price K.
Assuming no dividends, the cash payoff at time T is: $\Pi_T = S_T - K$

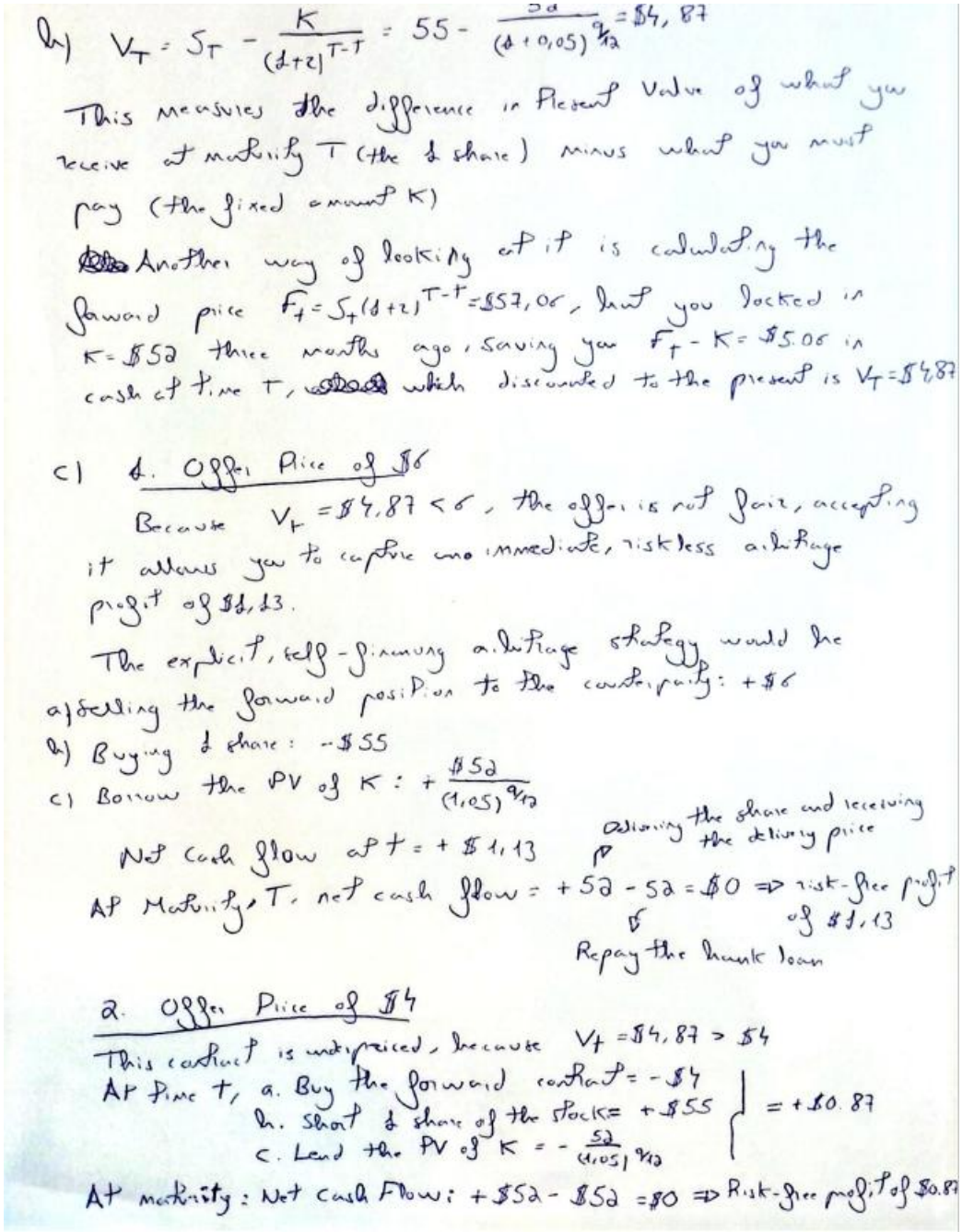
To replicate this exact payoff at maturity T, construct a portfolio P_t today using:
- Long 1 share of the underlying stock
- Borrow cash / short a zero-coupon risk-free bond

Using annual discrete compounding $(1+r)$, the amount borrowed at time t is $\Rightarrow Debt_t = \frac{K}{(1+r)^{T-t}}$

The net cost to construct this portfolio at time t is:

$$Cost(P_t) = S_t - \frac{K}{(1+r)^{T-t}}$$

And the value at time T will be $\Pi_T^P = S_T - K$, portfolio P provides an identical cash flow to the long forward contract across all possible outcomes of S_T.
By the Law of One Price (the principle of no-arbitrage), two financial positions that deliver identical payoffs in all future states must have the same value today $\Rightarrow V_t = S_t - \frac{K}{(1+r)^{T-t}}$



Question 2 — Replicating a Put and Recovering State Prices

Consider a one-period market with a stock currently priced at $S_0 = \$60$. Over the period, the stock either rises to \$75 (state u) or falls to \$48 (state d). A risk-free bond costs \$1 today and pays \$1.04 in either state (r = 4%).

(a) Find the portfolio (N shares of stock, \$B in the bond) that replicates a European put option with strike K = \$60 and hence find the put's no-arbitrage price today.

(b) Using the stock and bond prices, solve for the state prices q_u and q_d.

(c) Verify that pricing the put directly via Σ payoff$_i$ q_i gives the same value as your answer in (a). What feature of q_u and q_d guarantees, via the Fundamental Theorem of Asset Pricing, that this market admits no arbitrage?

② $S_0 = \$60$
$S_u = \$75$
$S_d = \$48$
$r = 4\%$

a) AP matching, the payoffs are

$P_u = \max(K - S_u, 0) = \max(60 - 75, 0) = \$0$

$P_d = \max(\$0 - 48, 0) = \$12$

A replicating portfolio consists of N shares of stock and B units of the bond

$$\begin{cases} u: & 75N + 1{,}04B = 0 \\ d: & 48N + 1{,}04B = 12 \end{cases}$$

$$27N = -12 \iff N = -\frac{12}{27} \approx -0{,}44 \text{ shares}$$

$$\Rightarrow 75\left(-\frac{4}{9}\right) + 1{,}04B = 0 \iff B = \frac{1250}{39} \approx \$32{,}05$$

The replicating portfolio consists of shorting $\frac{4}{9}$ shares of stock and buying \$32,05 of the risk free bonds

By the Law of one price, the no arbitrage cost of the put today is the initial cost of this portfolio:

$$P_0 = NS_0 + BB_0 = -\frac{4}{9} \cdot 60 + \frac{1250}{39} \cdot 1 = \$5{,}38$$

b) The states prices q_u and q_d are the current prices of Arrow-Debreu securities that pay \$1 in state u and d, respectively. They must price both the bond and the stock

b. $1{,}04\,q_u + 1{,}04\,q_d = 1 \Rightarrow q_u + q_d = \frac{1}{1{,}04} \iff q_u = \frac{1}{1{,}04} - q_u$

a. $75q_u + 48q_d = 60 \iff 75q_u + 48\left(\frac{1}{1{,}04} - q_u\right) = 60 \iff q_u = \frac{20}{39} \approx 0{,}51$

$\Rightarrow q_d = \frac{35}{78} \approx 0{,}45$

c) Evaluating the put option using the states prices:

$$P_0 = \sum_{i \in \{u,d\}} P_i q_i = 0\left(\frac{20}{39}\right) + 12\left(\frac{35}{78}\right) = \frac{70}{13} \approx \$5{,}38$$

According to the Fundamental Theorem of Asset Pricing, a market admits no arbitrage if and only if there exits a strictly positive state price vector consistent with observed asset prices:

$q_u = \frac{20}{39} > 0$ and $q_d = \frac{35}{78} > 0$

Because both states are strictly positive, it is mathematically impossible to form a zero-cost portfolio whose payoff is non-negative everywhere and strictly positive in at least one state.

Hence, the strict positivity of q_u and q_d guarantees that this market admits no arbitrage

Question 3 — A Third Asset

Consider the same two-state market as Question 2 (S_0 = \$60, rising to \$75 in state u or falling to \$48 in state d; r = 4%). A broker offers a claim that pays \$30 in state u and \$20 in state d, quoted at a market price of \$27.00.

Using the state prices found in Question 2, compute what this claim's price should be under no-arbitrage. Explain why the fact that this market is complete means you did not need to solve any new system of equations to answer this.

Do you agree to purchase this derivative from the broker?

(3) $S_0 = \$60$
$S_u = \$75$
$S_d = \$48$
$r = 4\%$

The no-arbitrage price of any contingent claim in this two-state market is the sum of its state payoffs weighted by the respective state prices:

$$V_0 = C_u q_u + C_d q_d = 30 \cdot \frac{20}{39} + 20 \cdot \frac{35}{78} = \$24{,}36$$

The no arbitrage price of the claim today is $24,36

A market is <u>complete</u> when every possible pattern of future payoffs across all states can be uniquely replicated by a portfolio of existing physical assets

- In this market → 2 future states (u and d)
  → 2 linearly independent assets (stock and risk-free bond)

$$\begin{pmatrix} 75 & 1{,}04 \\ 48 & 1{,}04 \end{pmatrix}$$

- Because the asset payoff matrix is square and invertible, there is a single, unique state price vector (q_u, q_d) consistent with the law of one price
- These state prices represent the universal market cost today of buying $1 delivered in each respective state, regardless of which instrument pays it.

q_u is the price of a contract that pays $1 if the market goes to state u, $0 otherwise, the same with q_d.

Once q_u and q_d are determined, every new derivative on these two states can be priced by taking the inner product of its payoffs with (q_u, q_d)

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No, you should not purchase the derivative from the broker, because it's overpriced: $27 - 24{,}36 = \$2{,}64 > 0$

Instead, you should sell (write) the claim to the broker for $27 and construct a replicating portfolio for $24,36